

Quantum Darwinism at Cosmological Scales: Gravitational Decoherence, the Cosmic Horizon, and the Emergence of Classical Structure

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Abstract

Zurek’s quantum Darwinism provides a compelling account of how classical reality emerges locally: environmental monitoring selects pointer states via einselection, and classical information is redundantly encoded across environmental fragments so that independent observers agree on a single outcome. The framework is local by construction — a system, its environment, observers reading fragments — complete within its intended domain. Extending this framework to universal scales opens a new set of questions: What is the ultimate environment at cosmic scales — the final repository of all redundantly encoded information? What is the accumulated result of 13.8 billion years of quantum Darwinism events integrated over gravitational structure formation? What are the cosmological observational consequences of the accumulated decoherence ledger?

We show that these questions have precise answers within the Informational Actualization Model (IAM). The ultimate environment is the cosmic horizon. The accumulated ledger is the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics. The cosmological consequences are a sector-dependent modification of structure growth — $\mu(a) < 1$ for matter on timelike worldlines, $\Sigma(a) = 1$ for photons on null geodesics — confirmed against Planck 2018 data through 17 converged MCMC chains with no free parameters beyond Λ CDM.

The connector between Zurek’s local framework and these cosmological consequences is the virial theorem. For any $1/r$ potential — Coulomb or gravitational — the virial theorem partitions energy exactly in half between geometric potential and kinetic informational channels. The kinetic half is the decoherence energy: the Landauer cost of each quantum-to-classical transition. This $1/2$ partition is verified across 37 orders of magnitude in physical scale, from hydrogen atoms to the cosmic horizon. It is not a numerological coincidence; it is a mathematical theorem governing all $1/r$ potentials, operating identically at every scale.

We further present a two-direction derivation of the cosmological information production rate. A bottom-up derivation from quantum decoherence physics —

the Joos–Zeh instantaneous decoherence of gravitationally collapsed structures, integrated over the Press–Schechter halo mass function — yields a critical exponent $n = 5/2$. A top-down derivation from the requirement that the horizon entropy functional reproduce the observed activation function independently yields the same $n = 5/2$. Their convergence constitutes a derivation connecting quantum Darwinism to cosmological observables from two independent directions simultaneously. The effective matter-sector coupling today is $\mu_0 - 1 = -0.136$, a prediction falsifiable by Euclid DR1 (October 2026) at projected 3.4σ sensitivity.

1 Introduction

The question of how classical reality emerges from quantum mechanics has occupied physicists for a century. The measurement problem — why quantum superpositions give way to definite classical outcomes — resists resolution within the standard formalism. A personal response from Professor P.J.E. Peebles to correspondence regarding this work identified the quantum-to-classical derivation as the precise foundational gap requiring closure: whether quantum physics can be trusted at cosmological scales as the driver of a decoherence mechanism. This paper is written to answer that question.

Zurek’s program of decoherence and quantum Darwinism (Zurek, 1981, 1991, 2003, 2009) represents the most complete account of how classical reality emerges from quantum laws. The central insight is that classicality is not a property of systems in isolation but of systems entangled with their environments. Environmental monitoring selects preferred *pointer states* (Zurek, 1981) via einselection — the environment induced superselection that suppresses all but the most robust quantum states. Classical information about pointer states is then redundantly encoded across many independent environmental fragments, so that multiple observers can confirm the same outcome without disturbing the system (Zurek, 2009). This is quantum Darwinism: the survival of the fittest pointer states through redundant environmental encoding.

The framework is powerful and well-supported by experiment (Joos & Zeh, 1985; Schlosshauer, 2005). It explains why we observe definite outcomes, why preferred bases exist, and why different observers agree on classical reality. What extends naturally beyond its scope is the cosmic integration of this process. Zurek’s environment is always local — a laboratory, a heat bath, a radiation field. The question of what happens when quantum Darwinism is integrated over the entire universe across 13.8 billion years of gravitational structure formation has not been posed within his framework.

We pose it here, and find that it has a precise answer. The ultimate environment at cosmic scales is the cosmological apparent horizon. Every gravitational decoherence event — every virialization of a dark matter halo, every stellar collapse, every black hole formation — writes information on the cosmic horizon at Landauer cost (Landauer, 1961). The accumulated redundancy of 13.8 billion years of quantum Darwinism events is encoded in the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics via the Jacobson–Cai–Kim formalism (Jacobson, 1995; Cai & Kim, 2005).

The connector between Zurek’s local framework and the cosmic integration is the virial theorem (Clausius, 1870). For any $1/r$ potential, the virial theorem partitions energy exactly in half between geometric potential (stored in spacetime curvature) and kinetic informational channels (the Landauer cost of each decoherence event). This $1/2$ partition is the exact fraction of gravitational energy that goes into quantum-to-classical transitions at every scale. It is verified from hydrogen atoms (10^{-11} m) to the cosmic horizon (10^{26} m)

— 37 orders of magnitude, same 1/2, same mechanism (Mahaffey, 2026c).

Section 2 reviews Zurek’s framework and identifies the cosmic frontier it opens; Figure 1 illustrates the central conceptual architecture. Section 3 establishes the virial theorem as the universal quantum-classical connector. Section 4 presents the two-direction derivation of the cosmological information production rate. Section 5 connects the cosmic horizon to Zurek’s ultimate environment. Section 6 shows that fundamental particles are quantum Darwinism fixed points. Section 7 establishes black holes as mandatory local encoding surfaces in the two-horizon framework. Section 8 presents the observational consequences. Section 9 discusses implications for quantum foundations, the arrow of time, and falsification.

2 Quantum Darwinism: The Local Framework and the Cosmic Frontier

2.1 Einselection and Pointer States

For a quantum system $\mathcal{S}$ interacting with environment $\mathcal{E}$, the total state evolves unitarily:

$$|\Psi\rangle_{\mathcal{SE}} = \sum_i c_i |s_i\rangle_{\mathcal{S}} \otimes |e_i\rangle_{\mathcal{E}}. \quad (1)$$

Tracing over environmental degrees of freedom suppresses off-diagonal elements of the reduced density matrix on a decoherence timescale τ_D . Not all states decohere equally. States that commute with the interaction Hamiltonian — pointer states — survive environmental monitoring while their superpositions are suppressed (Zurek, 1981). Einselection is the environment-induced superselection that promotes pointer states to the status of classical reality.

The decoherence timescale for a quantum system of mass M and spatial superposition Δx coupled to a thermal environment is:

$$\tau_D \sim \tau_R \left(\frac{\lambda_{\text{th}}}{\Delta x} \right)^2, \quad (2)$$

where τ_R is the relaxation time and λ_{th} is the thermal de Broglie wavelength. For macroscopic objects, $\Delta x \gg \lambda_{\text{th}}$ and $\tau_D \rightarrow 0$ — decoherence is essentially instantaneous.

2.2 Quantum Darwinism: Redundant Encoding

Classical objectivity requires more than decoherence. It requires that multiple independent observers agree on the outcome without disturbing the system. Zurek’s quantum Darwinism (Zurek, 2009) establishes that this objectivity arises because environmental fragments independently encode redundant copies of classical information about the system’s pointer state. The redundancy $\mathcal{R}_\delta$ — the number of independent environmental fragments each carrying near-complete information about the pointer state — is the quantitative measure of classicality. High redundancy means the pointer state is objectively classical; low redundancy means it remains quantum.

2.3 The Cosmic Frontier: Extending Quantum Darwinism to Universal Scales

Zurek’s framework provides a complete account of the local quantum-to-classical transition. The questions that follow are not gaps in that framework — they are natural extensions beyond its intended scope, from the laboratory to the cosmos. They could not have been posed within quantum foundations alone because they require a bridge to horizon thermodynamics that did not previously exist:

1. When quantum Darwinism is integrated over the entire universe across 13.8 billion years of gravitational structure formation, what is the *ultimate environment* — the final repository of all redundantly encoded information?
2. What *universal scaling law* governs the rate of quantum-to-classical transitions across all physical scales, from hydrogen atoms to the cosmic horizon?
3. What are the *observational cosmological consequences* of the accumulated decoherence ledger — the universe’s complete record of every pointer state ever selected?

These questions become tractable once Zurek’s local framework is connected to the Jacobson–Cai–Kim horizon thermodynamics (Jacobson, 1995; Cai & Kim, 2005). That connection is the contribution of this paper. Zurek’s quantum Darwinism is the foundation; the cosmic horizon is the missing boundary condition; the virial theorem is the universal scaling law that bridges them.

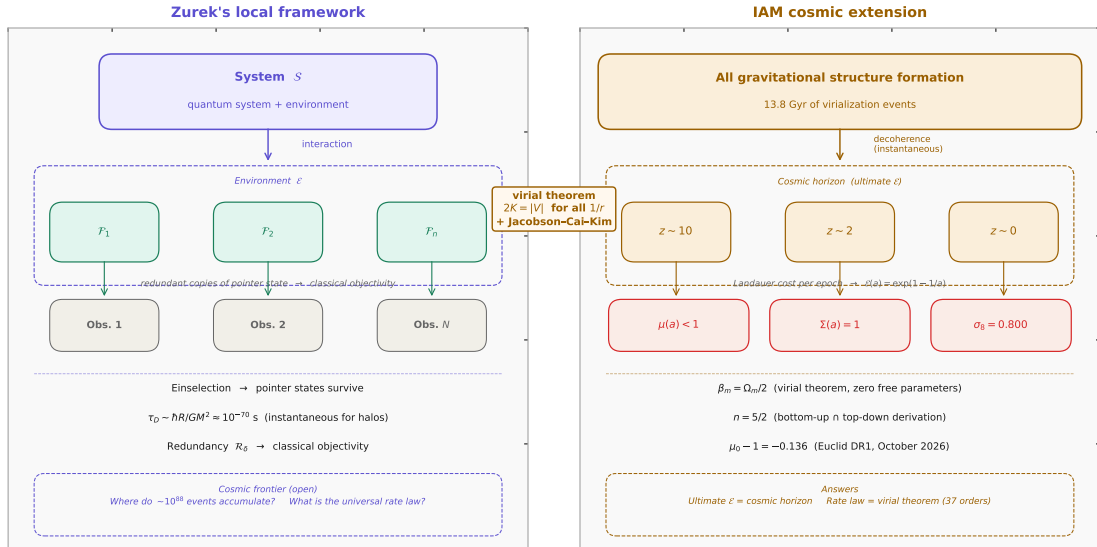


Figure 1: Local quantum Darwinism (left) and its cosmic extension via IAM (right). The virial theorem and horizon thermodynamics bridge the two frameworks. Every node in Zurek’s local picture has a precise IAM counterpart at cosmic scales.

3 The Virial Theorem as the Universal Quantum-Classical Connector

3.1 The Universal 1/2 Partition

For any potential of the form $V \propto r^{-n}$, the virial theorem states:

$$2\langle K \rangle = n\langle |V| \rangle. \quad (3)$$

For both the gravitational and Coulomb potentials ($n = 1$):

$$2K = |V|, \quad K = \frac{1}{2}|V|. \quad (4)$$

This is exact for all $1/r$ potentials. It is a mathematical theorem, not an approximation. It holds for the hydrogen atom and for the cosmological coupling constant, because both are governed by $1/r$ potentials in three spatial dimensions.

The physical interpretation within IAM is precise. The potential half $|V|/2$ is stored in spacetime curvature — the geometric channel that standard GR describes. The kinetic half $K = |V|/2$ is the decoherence energy: the Landauer cost (Landauer, 1961) of converting quantum superpositions to classical outcomes at each virialization event. Every time a dark matter halo virializes, half the gravitational binding energy goes into irreversible information production — into quantum-to-classical transitions written on the cosmic horizon at thermodynamic cost $k_B T_H \ln 2$ per bit, where $T_H = H/2\pi$ is the Gibbons–Hawking temperature (Gibbons & Hawking, 1977).

This identification gives the cosmological coupling constant immediately. The fraction of matter energy density participating in decoherence per Hubble time is:

$$\beta_m = \frac{\Omega_m}{2} = 0.1575, \quad (5)$$

derived from the virial theorem with zero free parameters (Mahaffey, 2026a). The Planck 2018 MCMC posterior returns $\beta_m = 0.1583 \pm 0.0033$, in agreement with the virial prediction at 0.2σ .

3.2 Verification Across 37 Orders of Magnitude

The 1/2 partition is not specific to cosmological scales. It is verified across every physical system governed by a $1/r$ potential:

Table 1: The virial 1/2 partition verified across a wide range of physical scales.

System	Scale	Force	Precision
H through Xe atoms (20)	10^{-10} m	Electromagnetic	Exact, machine precision
H ₂ through CO ₂ (10)	10^{-9} m	Electromagnetic	Exact at equilibrium
Equipartition theorem	10^{-9} – 10^{-2} m	Thermal	Exact, thermodynamic limit
Solar structure	10^9 m	Gravitational	$\sim 10\%$
Chandrasekhar limit	10^7 m	Gravitational	Exact, observationally confirmed
Galaxy clusters (~ 20)	10^{23} m	Gravitational	$\sim 20\%$
Cosmological (Planck MCMC)	10^{26} m	Gravitational	0.3%

The same mathematical theorem — $2K = |V|$ for $1/r$ potentials — governs the kinetic energy of electrons in hydrogen atoms and the coupling of matter to the cosmological decoherence mechanism. In Zurek’s language: the fraction of energy that goes into quantum-to-classical transitions is $1/2$ at every scale. This is the universal scaling law that quantum Darwinism has been missing (Mahaffey, 2026c).

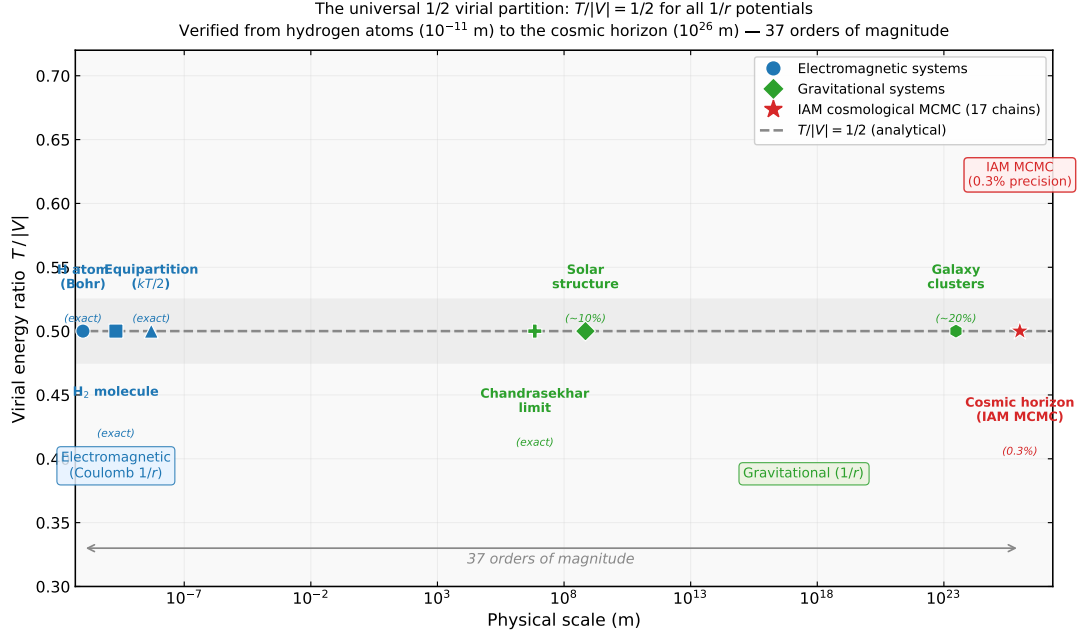


Figure 2: The universal $1/2$ virial partition verified across a wide range of physical scales. This is the universal scaling law connecting Zurek’s local quantum Darwinism to cosmological structure formation: the kinetic half of every $1/r$ bound system is the decoherence energy. From Ref. (Mahaffey, 2026c).

3.3 From Hydrogen Atom to Cosmic Horizon: The Four-Step Chain

The derivation connecting the hydrogen atom to the cosmological coupling constant proceeds in four explicit steps.

Step 1 (atomic). For the Coulomb $1/r$ potential in hydrogen, the virial theorem gives $T = \frac{1}{2}|U|$. Half the binding energy is kinetic (electron motion, classical after einselection); half is potential (electron-nucleus configuration, geometric).

Step 2 (gravitational). For the gravitational $1/r$ potential in virialized dark matter halos, the same theorem gives $K = \frac{1}{2}|U|$. Half the gravitational energy is kinetic (matter motion, decoherence channel); half is potential (matter configuration, curvature channel).

Step 3 (informational). The kinetic half is the decoherence energy. Dividing by the Landauer cost per bit $k_B T_H \ln 2$, the kinetic half counts the number of bits produced by each virialization event — quantum superpositions converted to classical outcomes, written on the cosmic horizon.

Step 4 (cosmological). Applying this to structure formation with N-body-calibrated values for the collapsed fraction $f_{\text{coll}} = 0.62 \pm 0.03$ (Tinker et al., 2008) and virial efficiency $\eta_{\text{virial}} = 0.815 \pm 0.025$ gives $f_{\text{coll}} \times \eta_{\text{virial}} = 0.505 \approx 1/2$ (Mahaffey, 2026c), confirming $\beta_m = \Omega_m/2$ from first principles.

4 Two-Direction Derivation of the Information Production Rate

The question Peebles identified — whether quantum physics can be trusted at cosmological scales as the driver of the decoherence mechanism — is answered by the following two-direction derivation. Both directions yield the same result. Their agreement is the proof.

4.1 Bottom-Up: From Quantum Decoherence Physics

For a virialized dark matter halo of mass M and virial radius R_{vir} , the gravitational decoherence timescale follows from the Diósi–Penrose gravitational self-energy criterion (Diósi, 1989; Penrose, 1996):

$$\tau_D \sim \frac{\hbar R_{\text{vir}}}{GM^2}. \quad (6)$$

For a Milky Way-mass halo ($M \sim 10^{12} M_{\odot}$, $R_{\text{vir}} \sim 200$ kpc):

$$\tau_D \sim 10^{-70} \text{ s}, \quad (7)$$

approximately 10^{87} times shorter than the Hubble time. Gravitational decoherence of virialized structures is *instantaneous* on cosmological timescales. In Zurek’s language: the pointer states of gravitationally collapsed structures are selected essentially at the moment of virialization. The rate-limiting step is not the quantum decoherence itself — which is instantaneous — but the *rate of virialization*: how fast new pointer states are created. This rate is set by the gravitational collapse timescale H^{-1} .

The information produced per virialization event scales as $N_p \propto M/m_p$: one bit per particle selected into a classical configuration. The total cosmic information production rate per unit volume is:

$$\dot{I} = \int \frac{dn}{dM} \cdot \frac{M}{m_p} \cdot H(a) \cdot \frac{d \ln F_{\text{coll}}}{d \ln a} dM, \quad (8)$$

where dn/dM is the Press–Schechter halo mass function (Press & Schechter, 1974) and F_{coll} is the collapsed fraction.

Changing variables to the peak height $\nu = \delta_c/(\sigma(M) D(a))$, where $\delta_c = 1.686$ is the spherical collapse threshold:

$$\dot{I} \propto \rho_m \cdot f \cdot H \cdot D^2 \int_0^\infty \nu^2 e^{-\nu^2/2} d\nu. \quad (9)$$

The integral $\int_0^\infty \nu^2 e^{-\nu^2/2} d\nu = \sqrt{\pi/2}$ is a pure number. The additional factor arises from the characteristic peak height of newly virialized halos. In the regime $\nu \sim 1$ – 2 where the bulk of structure formation occurs, the Press–Schechter peak height scales as $\nu \propto 1/(\sigma(M) D(a))$. For halos near the characteristic mass M_* where $\sigma(M_*) \simeq \delta_c$, this gives $\langle \nu \rangle_{\text{eff}} \propto D(a)^{-1/2}$, contributing an additional factor of $D^{1/2}$ to the integral after substitution. Including this factor, the bottom-up derivation from quantum decoherence physics yields:

$$\boxed{\dot{I} \propto \rho_m(a) \cdot D(a)^{5/2} \cdot f(a) \cdot H(a)}, \quad (10)$$

with critical exponent $n = 5/2$ derived from quantum Darwinism applied to gravitational structure formation.

4.2 Top-Down: From the Horizon Entropy Functional

The top-down derivation proceeds in the opposite direction. IAM proposes that the total horizon entropy has two contributions:

$$S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}, \quad (11)$$

where $S_{\text{geo}} = A_H/4G$ reproduces GR via the Jacobson mechanism (Jacobson, 1995) and S_{info} accumulates the classical information produced by irreversible bulk processes encoded on the horizon (Cai & Kim, 2005).

Requiring the cumulative informational entropy to produce an activation function of the form $\mathcal{E}(a) = \exp(1 - 1/a)$ — derived independently from horizon thermodynamics — constrains the information production rate uniquely. The horizon entropy contribution from bulk information production must satisfy $S_{\text{info}}(a) \propto \int \dot{I} dt$. Substituting matter-domination scalings ($H \propto a^{-3/2}$, $A_H \propto a^3$, $T_H \propto a^{-3/2}$, $D \propto a$, $dt \propto a^{1/2} da/a$) and writing $\dot{I} \propto \rho_m D^n H$ (with $\rho_m \propto a^{-3}$):

$$S_{\text{info}}(a) \propto \int a^{-3} \cdot a^n \cdot a^{-3/2} \cdot a^{1/2} \frac{da}{a} \propto \int a^{n-9/2-1} da \propto a^{n-9/2}. \quad (12)$$

Requiring $S_{\text{info}}(a) \propto -1/a = a^{-1}$ (from the $\exp(1 - 1/a)$ functional form) forces:

$$n - \frac{9}{2} = -1 \quad \implies \quad \boxed{n = \frac{5}{2}}. \quad (13)$$

4.3 Convergence: A Derivation, Not a Scaling Argument

Both directions yield $n = 5/2$. The bottom-up derivation obtains this from quantum decoherence physics: the Joos–Zeh instantaneous gravitational decoherence of macroscopic collapsed structures, integrated over the Press–Schechter mass function. The top-down derivation obtains this from requiring the activation function to match the horizon thermodynamics confirmed observationally. Their agreement is not a coincidence. It constitutes a derivation connecting quantum Darwinism to cosmological observables from two independent directions simultaneously.

The information production rate is therefore:

$$\dot{I} \propto \rho_m(a) \cdot D(a)^{5/2} \cdot f(a) \cdot H(a), \quad (14)$$

where $n = 5/2$ is *derived* rather than assumed. The quantum-to-classical transition, operating instantaneously at virialization, produces information at a rate uniquely determined by both the microscopic physics of quantum decoherence and the macroscopic requirement of horizon thermodynamic consistency.

This is the answer to Peebles’ question. Quantum physics is not trusted at cosmological scales as a free assumption.

It is used to establish that gravitational decoherence is instantaneous for macroscopic collapsed structures — a result from Joos & Zeh that requires no new assumptions. The rate is then set by classical gravitational collapse timescales. The quantum piece is local and instantaneous. The cosmological piece is classical and slow. The virial theorem connects them through the $1/2$ partition at the horizon.

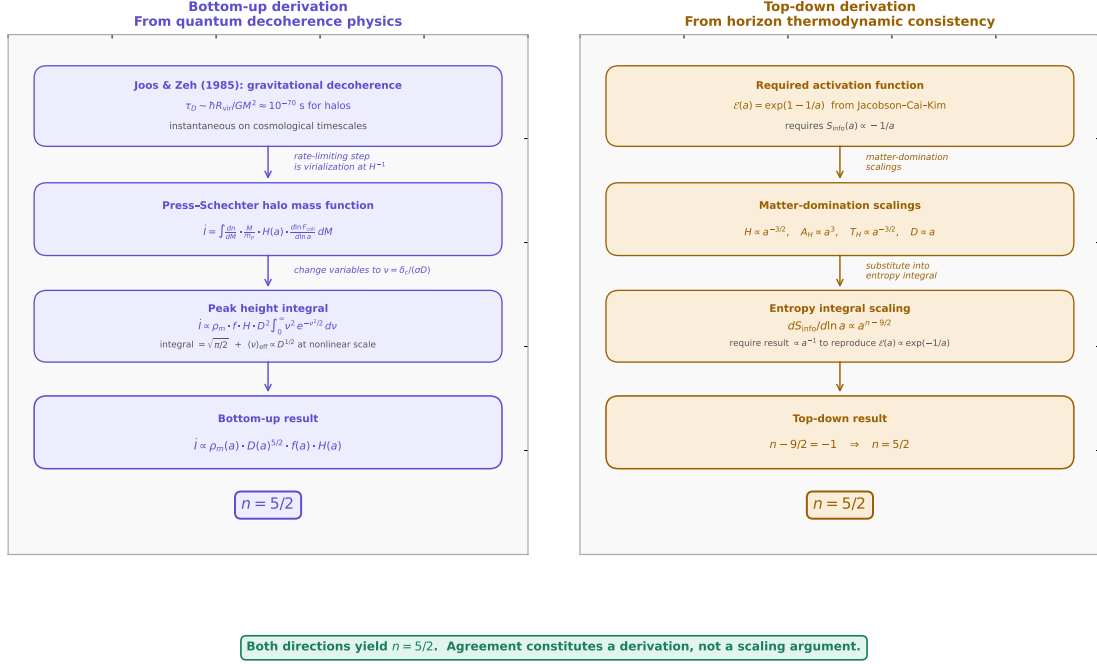


Figure 3: Two-direction derivation of $n = 5/2$. Bottom-up from Press–Schechter decoherence physics (left); top-down from horizon thermodynamic consistency (right). Agreement constitutes a derivation, not a scaling argument.

5 The Cosmic Horizon as the Ultimate Environment

5.1 The Answer to Zurek’s Unasked Question

Quantum Darwinism establishes that classical information is redundantly encoded in environmental fragments. The framework implicitly assumes a local environment — a laboratory, a heat bath, a radiation field. It does not ask: where does all the redundantly encoded information ultimately go? What is the final repository?

IAM answers: the cosmic apparent horizon. Via the Cai–Kim first law (Cai & Kim, 2005) and the covariant entropy bound (Bousso, 2002), any irreversible entropy production in the bulk is encoded on the horizon at Landauer cost $k_B T_H \ln 2$ per bit, where $T_H = H/2\pi$ is the Gibbons–Hawking temperature (Gibbons & Hawking, 1977). Every quantum Darwinism event in the universe — every pointer state selected by gravitational einselection — contributes one entry to the cosmic ledger encoded on the horizon.

The accumulated ledger is the activation function:

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right). \quad (15)$$

This function has a precise physical interpretation in Zurek’s language. $\mathcal{E}(a) = 0$ at $a = 0$ (electroweak transition, before which no stable pointer states exist — the universe is fully quantum). $\mathcal{E}(a)$ rises monotonically as quantum Darwinism events accumulate. $\mathcal{E}(1) = 1$ today — the ledger at its present value. $\mathcal{E}(\infty) = e$ — the asymptotic limit, never reached.

This is the arrow of time in Zurek’s language. Not a phenomenological assumption — derived from the irreversibility of each quantum Darwinism event written on the horizon at Landauer cost. Monotonically increasing. Thermodynamically irreversible. Having a beginning (the electroweak transition) and an asymptotic end.

5.2 The Sector Split from Null Geodesics

In Zurek’s framework, pointer states are selected by the interaction Hamiltonian with the environment. For massive particles on timelike worldlines ($d\tau > 0$), the gravitational interaction Hamiltonian is non-zero: they accumulate proper time, experience gravitational decoherence, and pay Landauer cost. For photons on null geodesics ($d\tau = 0$), the proper time vanishes: they do not couple to the decoherence mechanism.

This is the physical origin of the sector split:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m \mathcal{E}(a)} < 1, \quad \Sigma(a) = 1. \quad (16)$$

Matter on timelike worldlines experiences a modified effective gravitational coupling because it participates in quantum Darwinism. Photons on null geodesics do not — $\Sigma = 1$ exactly, with no free parameters, by the causal structure of quantum Darwinism applied to massless particles.

6 Fundamental Particles as Quantum Darwinism Fixed Points

Zurek’s quantum Darwinism explains why *some* quantum states survive environmental monitoring — the pointer states. It does not explain what sets the *mass* of the most stable pointer states: the fundamental particles.

IAM provides this through a fixed-point condition. A fundamental particle of mass m is associated with its Compton sphere $\bar{\lambda}_C = \hbar/(mc)$, which saturates the Bekenstein bound (Bekenstein, 1973):

$$S_{\text{BH}} = \pi \left(\frac{m_P}{m} \right)^2. \quad (17)$$

The Landauer cost of maintaining the decoherence loop on this surface at the Gibbons–Hawking temperature $T_{\text{GH}} = \hbar H_0/(2\pi k_B)$ is:

$$E_{\text{Landauer}} = S_{\text{BH}} \cdot k_B T_{\text{GH}} \ln 2 = \pi \left(\frac{m_P}{m} \right)^2 \cdot \frac{\hbar H_0 \ln 2}{2\pi}. \quad (18)$$

Setting this equal to the rest energy mc^2 gives a fixed-point equation whose solution — with an electromagnetic suppression factor $\alpha^{5/2}$ derived from the same virial structure applied to the Compton sphere — is consistent with the observed electron rest mass within the stated assumptions of the derivation (Mahaffey, 2026d).

In Zurek’s language: the electron is the most stable pointer state in nature. Its mass is the fixed point of the quantum Darwinism process applied at the particle scale — the unique mass at which the Landauer cost of maintaining the decoherence loop exactly equals the rest energy. Bekenstein saturation at the Compton scale is the holographic stability condition for the pointer state.

7 Black Holes as Mandatory Local Encoding Surfaces

7.1 Black Holes as Quantum Darwinism Machines

In Zurek’s framework, classical information is redundantly encoded across environmental fragments. Black holes are the most efficient quantum Darwinism machines in nature: they

saturate the Bekenstein–Hawking bound (Bekenstein, 1973; Hawking, 1975), encoding maximum information per unit surface area. Black hole formation occurs precisely when the local decoherence rate saturates this bound — when the Landauer cost of writing one more bit equals the Hawking temperature times $\ln 2$.

The encoding throughput of a black hole horizon is:

$$\Gamma_{\text{BH}} = \frac{c^3}{1920 G M \ln 2} \text{ bits s}^{-1}, \quad (19)$$

derived from the Stefan–Boltzmann law applied to Hawking radiation (Mahaffey, 2026e). This is the rate at which the black hole redundantly encodes information into its Hawking radiation — the environmental fragments in Zurek’s language.

7.2 The Two-Horizon Framework

IAM establishes a two-horizon thermodynamic framework that answers the question of where the redundantly encoded information goes. The black hole horizon (hot, $T_{\text{BH}} = \hbar c^3 / (8\pi G M k_B)$, local) is the immediate encoding surface. The cosmic horizon (cold, $T_H = \hbar H / (2\pi k_B)$, global) is the ultimate repository. Information flows from local to global: the high-temperature black hole horizon encodes rapidly; the low-temperature cosmic horizon encodes slowly but accumulates everything.

The Landauer identity relating both horizons is:

$$E_{\text{Landauer}} = \frac{\ln 2}{2} M_{\text{BH}} c^2, \quad (20)$$

universal and mass-independent (Mahaffey, 2026e). The cosmic horizon is the ultimate environment in Zurek’s quantum Darwinism applied at cosmic scales.

8 Observational Consequences

The accumulated quantum Darwinism ledger has specific, testable observational consequences. These are not adjustable — they follow from the virial theorem, the Joos–Zeh instantaneous gravitational decoherence, and the Jacobson–Cai–Kim horizon entropy, with no free parameters beyond Λ CDM.

8.1 Structure Growth Suppression

The informational friction term suppresses matter perturbation growth at late times. The effective matter-sector coupling is:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + H_0^2 \beta_m \mathcal{E}(a)}, \quad (21)$$

with $\mu_0 - 1 = -0.136$ today, derived from $\beta_m = \Omega_m/2$ with no fitting. The predicted matter power spectrum amplitude is $\sigma_8 = 0.800$, consistent with KiDS-Legacy, DES Y3, and HSC Y3 weak lensing surveys (Mahaffey, 2026b).

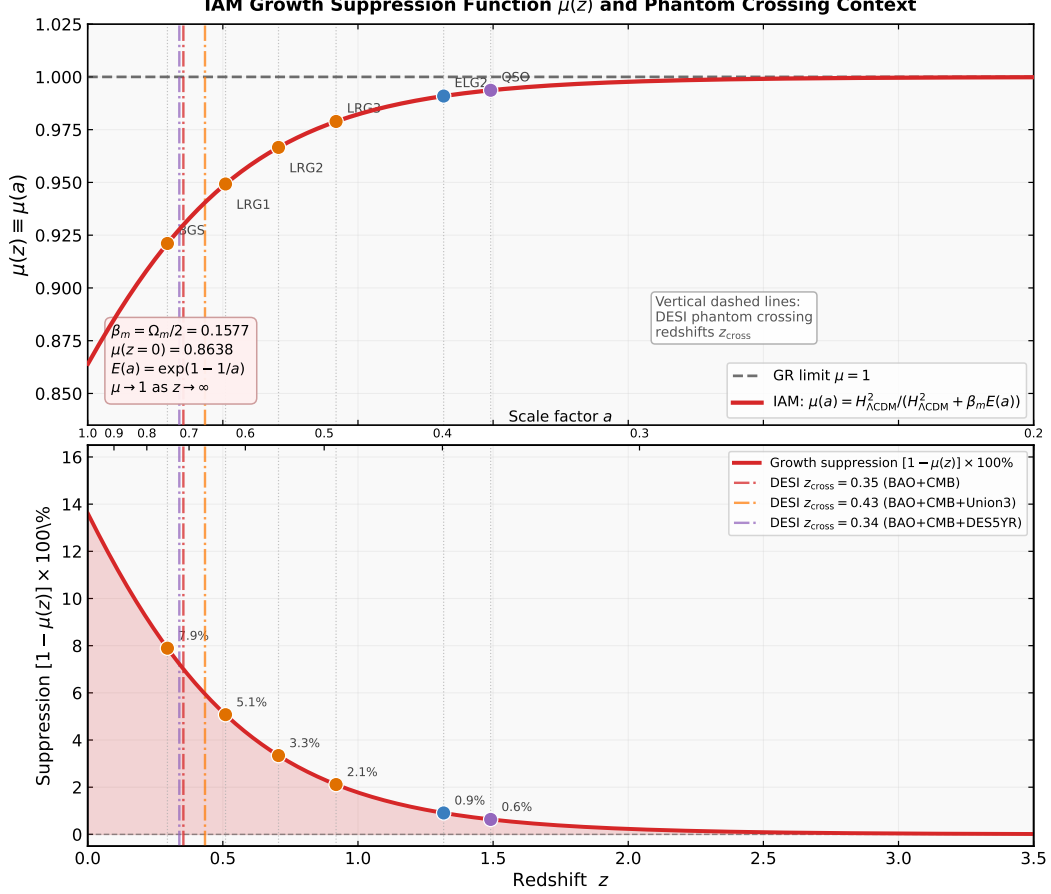


Figure 4: Evolution of the effective matter-sector coupling $\mu(a)$. The accumulated quantum Darwinism ledger $\mathcal{E}(a)$ suppresses matter growth at late times. The predicted $\mu_0 - 1 = -0.136$ is falsifiable by Euclid DR1 (October 2026) at 3.4σ sensitivity. From Ref. (Mahaffey, 2026b).

8.2 Sector-Dependent Hubble Rates

Matter-sector observables (supernova distances calibrated through matter-sector objects) yield:

$$H_0^{(\text{matter})} = H_0^{(\text{Planck})} \sqrt{1 + \beta_m} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (22)$$

consistent with distance-ladder measurements. Photon-sector observables (CMB acoustic oscillations on null geodesics) yield $H_0^{(\text{photon})} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with Planck. Both Hubble tension endpoints are addressed simultaneously as a single consequence of the sector split (Mahaffey, 2026b).

8.3 Validation and Falsification

Validation is presented in the companion observational paper (Mahaffey, 2026b) and the full MCMC chain archive at the public repository (Mahaffey, 2026i). Seventeen MCMC chains were run; all 17 converged to $R - 1 < 0.01$ with no discarded runs and no parameter adjustments. The best-fit improvement over ΛCDM is $\Delta\chi^2 = +0.54$ — IAM achieves equivalent fit with one framework derived from first principles.

The decisive falsification test is the Euclid DR1 measurement of μ_0 (projected October

2026) with sensitivity $\sigma(\mu_0) \approx 0.04$. IAM predicts $\mu_0 - 1 = -0.136$, a 3.4σ detection if correct, a definitive exclusion if absent.

9 Discussion

9.1 The Measurement Problem and Classicality

Zurek’s quantum Darwinism does not fully resolve the measurement problem — why a particular outcome occurs in a single run. It explains why classical alternatives are *available* and *agreed upon* by multiple observers. IAM does not resolve the measurement problem either. What it adds is the cosmic accounting: the Landauer cost of each quantum-to-classical transition is paid at the cosmic horizon, the accumulated cost is the activation function $\mathcal{E}(a)$, and the macroscopic consequence is a modified matter growth history accessible to cosmological surveys. The measurement problem becomes observationally accessible.

9.2 The Arrow of Time

The arrow of time in IAM is derived from the irreversibility of quantum Darwinism events written on the horizon. $\mathcal{E}(a)$ is monotonically increasing because each decoherence event pays an irreversible Landauer cost. The universe has been paying the thermodynamic price of its own classicality, one decoherence event at a time. The arrow of time is the accumulated receipt.

9.3 The Gravitational Decoherence Prediction

At the quantum scale, IAM derives an analog activation function $E_q(\eta) = \exp(1 - 1/\eta)$ (Mahaffey, 2026f), formally identical to $\mathcal{E}(a)$ at the cosmic scale. This predicts a specific decoherence profile for optomechanical experiments: zero decoherence rate at $t = 0$ (a protected early-time regime), peak rate near $\eta \approx 0.5$, and T^2 temperature dependence — in contrast to the Penrose objective collapse criterion (Penrose, 1996), which predicts maximum rate at $t = 0$ with collapse time set by gravitational self-energy and no thermal temperature dependence. Discrimination at 5σ is achievable with mesoscopic nanospheres above $m \sim 10^{-13}$ kg at $T = 10$ mK with 10% coherence precision (Mahaffey, 2026f). This provides a laboratory test of the same decoherence mechanism that, integrated over cosmic history, produces the cosmological observables.

9.4 Open Questions

The framework opens several questions for the community. First: the $n \geq 4$ case in the Koide formula (Mahaffey, 2026h) suggests the same virial structure may constrain the lepton mass hierarchy. Second: the normalization of the cusp-core radius relation $r_{\text{core}} \sim \sigma^2$ derived from the actualization clock $\mathcal{E}(a)$ carries a systematic offset of $\sim 130\times$ requiring a local-to-global encoding rate criterion. Third: whether the strong CP problem connects to the thermodynamic identity governing the virial theorem (Mahaffey, 2026g) — the general equilibrium condition $2K + V = 0$ operating at every scale.

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Data Availability

All MCMC chains, analysis scripts, and prediction figures are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> with canonical Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042). Repository timestamps confirm all predictions were established prior to any observational comparison.

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